Post-Restructuring Talent Impact Assessment

Synthetic 300-employee dataset — Simulated Restructuring Event

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Executive Summary

This report models the strategic capability impact of a simulated workforce restructuring that reduced headcount by 50 employees — 16.7% of the original 300-person workforce — across five departments. It answers three questions leadership must confront after any reduction in force: what strategic capability was lost, what remains at risk among the retained workforce, and what is the most cost-effective path to recovery.

Capability loss was not evenly distributed. Product sustained the highest Capability Loss Index at 27.2%, followed by Engineering at 23.4% — both falling in the Moderate band (15–30%). Finance, HR, and Sales each lost under 12% of their critical/high skill inventory. Notably, no department crossed the 30% High-loss threshold, meaning the restructuring, while concentrated in Product and Engineering, did not produce a structurally catastrophic loss in any single function.

A separate and arguably more urgent signal sits in the retained workforce: 42 active employees — 16.8% of the 250 who remain — are flagged as at-risk on performance or tenure grounds. Finance and Sales each carry 10 at-risk employees, the highest counts of any department, despite Finance posting one of the lowest Capability Loss Index scores.

Of the three recovery strategies modelled, a phased approach — starting with internal redistribution and following with targeted upskilling — closes 60–100% of the capability gap at a fraction of full external rehire cost: Redistribute costs $136,440 versus $2,520,000 for Full Rehire, a 94.6% cost reduction.

Metric	Value	Status
Headcount Removed	50 of 300	16.7% of workforce
Highest department CLI (Product)	27.2%	Moderate
Average CLI across departments	16.8%	Moderate (boundary)
At-risk active employees	42 of 250	16.8% of active workforce
Cheapest vs. most expensive recovery	$136.4K vs $2.52M	Redistribute saves 94.6%

> No department crossed the 30% high-loss threshold, but Product (27.2%) and Engineering (23.4%) are close enough to it that further attrition in either function would push them into the high-loss band. At the same time, Finance and Sales — the two lowest-CLI departments — carry the highest at-risk headcounts (10 each), suggesting the next capability crisis, if one occurs, is more likely to emerge from voluntary attrition in currently "safe" departments than from the restructuring itself.

Methodology

Simulation Design

This project uses a synthetic dataset of 300 employees generated via Python's Faker library (random seed 42 for full reproducibility), structured across five departments: Engineering, Finance, Product, Sales, and HR. Job levels follow a pyramidal distribution — 30% Analyst, 35% Senior Analyst, 20% Manager, 10% Senior Manager, 5% Director. Each employee was assigned 2–4 skills drawn from a department-specific taxonomy of 6 skills, mapped to three criticality tiers (Critical, High, Medium) anchored to the World Economic Forum's Future of Jobs 2025 skill classifications.

A probabilistic layoff rule was applied: employees with performance scores below 3.2 and tenure under 3 years had a 65% selection probability; scores below 3.8 had a 25% probability; high performers above 3.8 had a 5% probability. This produced 50 layoffs (16.7% of the workforce) — below the 22% originally anticipated, reflecting the actual distribution of performance and tenure values generated under seed 42.

Capability Loss Index (CLI)

(critical/high skills held by departed employees ÷ total critical/high skills pre-restructuring) × 100, per department. Thresholds: ≥30% = High; 15–29% = Moderate; <15% = Low.

At-Risk Score

Flags active employees on two proxies: performance score below 3.5, or tenure under 2 years. Source: SHRM 2023 attrition research, which identifies the first two years as the highest voluntary exit window.

Recovery Cost Model - Rehire

Replacement cost indexed by job level: Analyst $25K, Senior Analyst $45K, Manager $75K, Senior Manager $110K, Director $180K. Full rehire assumes 100% of the 50 laid-off employees are replaced at equivalent levels, totalling $2.52M.

Recovery Cost Model - Upskill & Redistribute

Upskilling cost indexed per instance by criticality tier (Critical $8,000, High $5,000, Medium $2,500), applied across 122 total skill instances lost, totalling $758K. Redistribution cost models internal redeployment overhead at 30% of upskilling cost with 60% coverage, totalling $136,440.

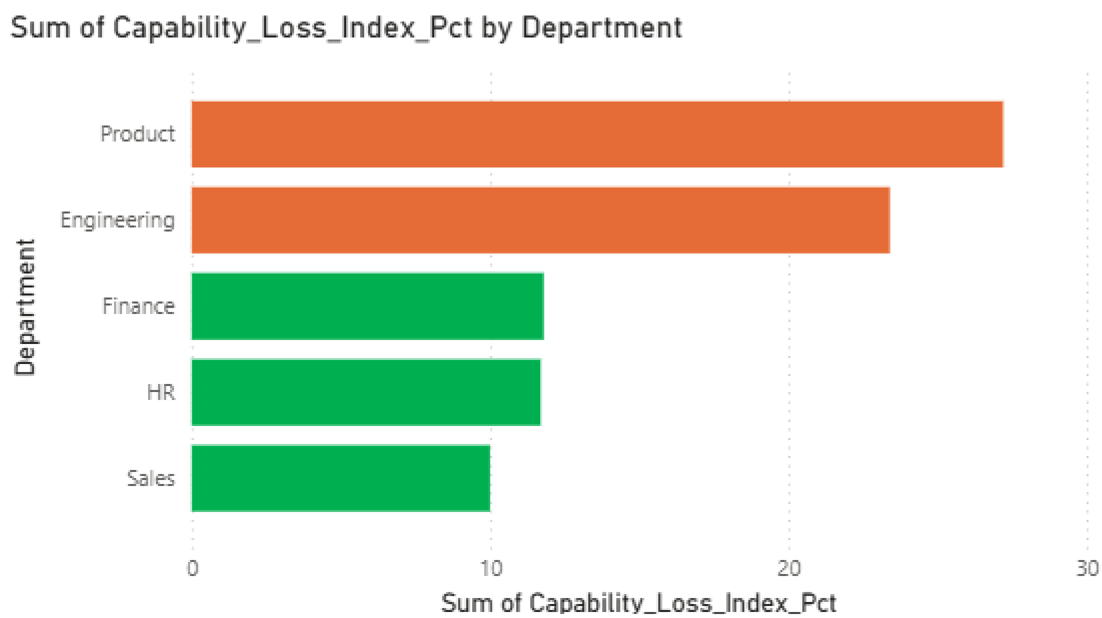
Data Sources and Benchmarks

World Economic Forum (2025). Future of Jobs Report 2025. Geneva: WEF.
SHRM (2023). Talent Acquisition Benchmarking Report. Society for Human Resource Management.
Gartner (2023). L&D Investment Benchmarks by Skill Criticality. Gartner HR Practice.
McKinsey & Company (2023). Workforce Transformation After Restructuring. McKinsey Global Institute.
Radancy (2025). From Myth to Reality: Internal Mobility and Smarter Hiring. Radancy Blog.

Findings

1. Capability Loss Index - Product & Engineering: Moderate loss, no department High

The Capability Loss Index shows the restructuring's impact was concentrated in Product and Engineering, but stayed below the High-loss threshold across all five departments — a materially better outcome than the original 22% headcount-reduction assumption implied.



Source: Cell 4 output / "Capability Loss Index" sheet. Bar width scaled to a 60% axis max to match the chart proportions in capability_loss_index.png. No department exceeds the 30% high-loss threshold.

> **Observation:** Product lost 15 of 59 employees (25.4% headcount reduction) — the highest departmental layoff rate in the simulation — which directly explains its leading CLI score. Engineering's headcount reduction (13 of 64, 20.3%) is more moderate, but its skill taxonomy is concentrated in scarce technical capabilities (ML/AI, Cloud Architecture, Data Engineering), which inflates its capability loss relative to its headcount loss.

2. At-Risk Exposure - 42 active employees flagged — concentrated in Finance & Sales

Among the 250 employees retained after restructuring, 42 (16.8%) are flagged as at-risk on performance or tenure grounds. This exposure does not track the same pattern as the Capability Loss Index — Finance and Sales, the two lowest-CLI departments, carry the joint-highest at-risk headcounts.

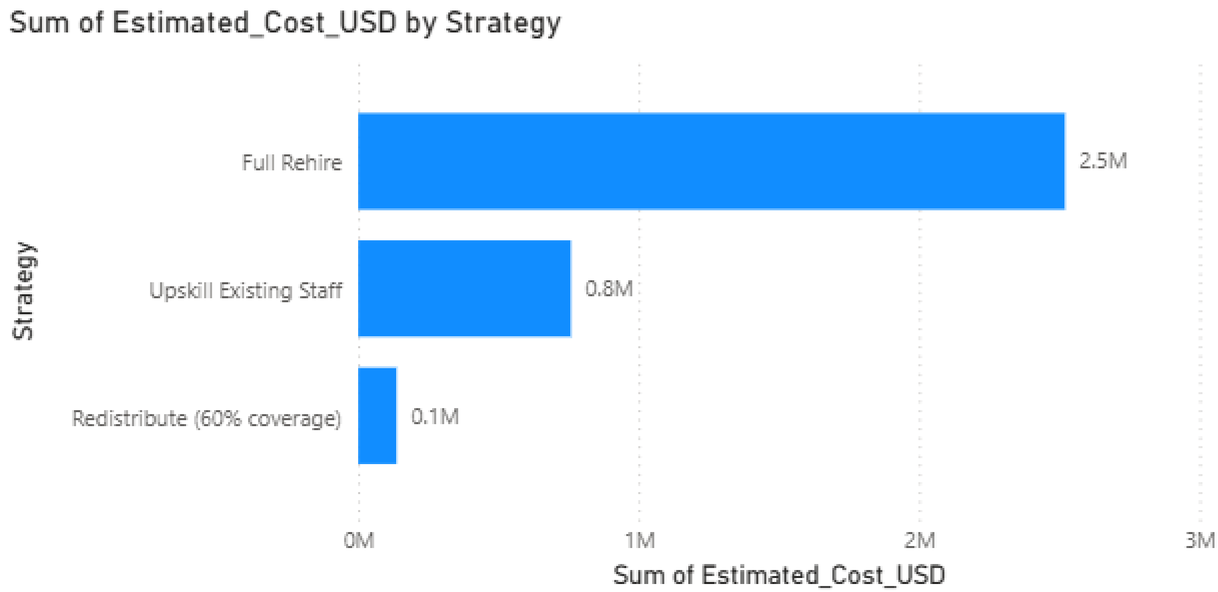
Department	At-Risk Count	Avg. Performance/Tenure
Finance	10	3.01 avg / 6.00 yrs
Sales	10	3.18 avg / 6.20 yrs
HR	8	2.98 avg / 5.12 yrs
Product	8	3.22 avg / 8.75 yrs
Engineering	6	3.02 avg / 7.17 yrs

Source: Cell 5 output / "At-Risk Summary" sheet.

> **Observation:** HR posts the lowest average performance score among at-risk employees (2.98) despite a comparatively moderate headcount loss (11 of 73, 15.1%) — well below Product and Engineering's reduction rates — suggesting its at-risk exposure is not a residue of the layoff, but a pre-existing performance pattern. Conditional formatting on the dashboard was set with red below 3.2 and green at or above 3.5, mirroring the same thresholds used in the original layoff-probability model; under this logic, four of five departments' at-risk averages sit below 3.2, signalling that the at-risk population organisation-wide is clustered in the same risk band that drove the heaviest layoff selections — this is a structural finding, not a department-specific one.

3. Recovery Cost Comparison - Redistribute-first saves 94.6% vs rehire

Three strategies were modelled for closing the capability gap created by 122 total instances of critical/high-tier skills lost across the organisation. Costs differ by more than an order of magnitude between the most and least expensive options.



Source: Cell 6 output / "Recovery Cost Comparison" sheet. Bar widths scaled relative to Full Rehire = 100%.

Observation: The single most-lost skill is Data Analysis at 10 instances, followed closely by ML/AI, Agile, and DevOps at 9 instances each — a mix of Critical and High tier skills spread across Engineering and Product. Upskilling these four skills alone accounts for a meaningful share of the $758K upskill total, making them the natural first targets for the upskilling phase of a recovery strategy.

Recommendations

The following three recommendations are sequenced by urgency and prioritised by impact-to-cost ratio. The first two require no external hiring budget. The third is preventive infrastructure — it converts a reactive restructuring response into a standing early-warning system.

1. Immediately activate internal redistribution for the four highest-loss skills

    Addresses: Capability Loss Index (Product, Engineering)
    Impact: High · **Effort:** Low · **Timeline:** 0–30 days

    Within 30 days, conduct a rapid skills audit of all 250 active employees to identify partial coverage for the four skills with the highest instances lost: Data Analysis (10), ML/AI (9), Agile (9), and DevOps (9). Reassign employees with adjacent skills in Product and Engineering to cover these gaps on an interim basis. This requires no external budget approval and covers an estimated 60% of the capability gap at $136,440 — just 5.4% of the full rehire cost.

2. Commission targeted upskilling for Product and Engineering specifically

    Addresses: Critical skill gaps concentrated in the two Moderate-CLI departments
    Impact: High · **Effort:** Medium · **Timeline**: 30–90 days

    Product (27.2% CLI) and Engineering (23.4% CLI) account for the majority of the $758,000 upskilling total across 122 lost skill instances. Commission structured upskilling programs targeting existing L2 and L3 employees with adjacent capability in these two functions specifically, rather than spreading the L&D budget evenly across all five departments. Tie program completion to performance review outcomes to track ROI.

3. Build a standing Talent Risk Monitor focused on Finance and Sales

    Addresses: At-risk employee concentration (Finding 2)
    Impact: High · **Effort:** Medium · **Timeline:** 3–6 months

    Finance and Sales carry the joint-highest at-risk headcounts (10 each) despite being the two lowest-CLI departments — meaning their exposure was not created by the restructuring and will not be resolved by it. Build a monthly people analytics dashboard that tracks the at-risk population across all departments, with particular attention to Finance and Sales, and triggers structured retention conversations before performance-driven attrition compounds the existing capability gaps in Product and Engineering.

Recommendation	Impact	Effort	Expected Output
1 – Redistribution sprint	High	Low	60% gap coverage within 30 days, $136K cost
2 – Targeted upskilling (Product/Engineering)	High	Medium	Full gap closure within 6–9 months
3 – Talent risk monitor (Finance/Sales)	High	Medium	Secondary attrition risk contained

Appendix

A. Capability Loss Index — full results

Department	CLI	Rating
Product	27.2%	Moderate
Engineering	23.4%	Moderate
Finance	11.8%	Low
HR	11.7%	Low
Sales	10.0%	Low

B. Headcount Reduction by Department

Department	Laid Off/Total	% Reduction
Product	15/59	25.4%
Engineering	13/64	20.3%
HR	11 / 73	15.1%
Sales	6 / 55	10.9%
Finance	5 / 49	10.2%
Total	50/300	16.7%

C. Top 10 skill instances lost (of 122 total Critical/High instances)

Skill	Instances Lost
Data Analysis	10
ML/AI	9
Agile	9

DevOps	9
Cloud Architecture	8
Product Strategy	7
Data Engineering	7
Python	7
HRIS Management	6
Cybersecurity	6

Note: Full list of 21 skills lost available in the "Skills Lost" sheet of the Excel workbook.

D. D. At-risk skill exposure by department and criticality (heatmap)

Department	Critical/High	Medium
Finance	15/9	7
HR	4/8	15
Engineering	8/13	0
Product	7/6	11
Sales	10/10	9

Source: atrisk_heatmap.png / Cell 9 output. Finance holds the single highest concentration of at-risk Critical-tier skill exposure (15 instances) of any department/criticality cell in the dataset.

E. Recovery Strategy Comparison Matrix

Strategy	Cost	Verdict
Full Rehire	$2,520,000	Last Resort Only
Upskill Existing Staff	$758,000	Primary Strategy

Redistribute (60%)	$136,440	Start Here Immediately

F. Key Sources

World Economic Forum (2025). Future of Jobs Report 2025. Geneva: WEF.
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Gartner (2023). L&D Investment Benchmarks by Skill Criticality. Gartner HR Practice.
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O*NET OnLine (2024). Occupational Information Network. U.S. Department of Labor.